

4.8.15

AI25BTECH11019 - MENAVATH SAI SANJANA

Question:

A line l passes through point $(-1, 3, -2)$ and is perpendicular to both the lines

$$\frac{x-1}{1} = \frac{y-1}{2} = \frac{z+1}{3}, \quad \frac{x+2}{-3} = \frac{y}{2} = \frac{z}{5}. \quad (0.1)$$

Find the vector equation of the line l and its distance from the origin.

Solution:

Let the direction vector of the required line be

$$\mathbf{d} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}. \quad (0.2)$$

Since the line is perpendicular to the two given lines, the direction vector satisfies

$$\mathbf{d}_1^\top \mathbf{d} = 0 \Rightarrow 1 \cdot a + 2 \cdot b + 3 \cdot c = 0, \quad (0.3)$$

$$\mathbf{d}_2^\top \mathbf{d} = 0 \Rightarrow -3 \cdot a + 2 \cdot b + 5 \cdot c = 0. \quad (0.4)$$

Solve the system:

$$8b + 14c = 0 \Rightarrow b = -\frac{7}{4}c, \quad (0.5)$$

$$a + 2b + 3c = 0 \Rightarrow a = \frac{1}{2}c. \quad (0.6)$$

Choosing $c = 4$ to get integer values:

$$\mathbf{d} = \begin{pmatrix} 2 \\ -7 \\ 4 \end{pmatrix}. \quad (0.7)$$

Vector equation of the line through $P = (-1, 3, -2)$:

$$\mathbf{r} = \begin{pmatrix} -1 \\ 3 \\ -2 \end{pmatrix} + t \begin{pmatrix} 2 \\ -7 \\ 4 \end{pmatrix}, \quad t \in \mathbb{R}. \quad (0.8)$$

Distance from origin:

$$t = -\frac{\mathbf{P} \cdot \mathbf{d}}{\mathbf{d} \cdot \mathbf{d}} = -\frac{(-1)(2) + 3(-7) + (-2)(4)}{2^2 + (-7)^2 + 4^2} = \frac{31}{69}, \quad (0.9)$$

$$\mathbf{Q} = \mathbf{P} + t\mathbf{d} = \begin{pmatrix} -\frac{7}{69} \\ -\frac{10}{69} \\ -\frac{14}{69} \end{pmatrix}, \quad (0.10)$$

$$D = \|\mathbf{Q}\| = \sqrt{\frac{345}{69^2}} = \sqrt{\frac{5}{69}}. \quad (0.11)$$

Conclusion: The required line is

$$\mathbf{r} = \begin{pmatrix} -1 \\ 3 \\ -2 \end{pmatrix} + t \begin{pmatrix} 2 \\ -7 \\ 4 \end{pmatrix}, \quad t \in \mathbb{R}, \quad (0.12)$$

and its distance from the origin is

$$D = \sqrt{\frac{5}{69}}. \quad (0.13)$$

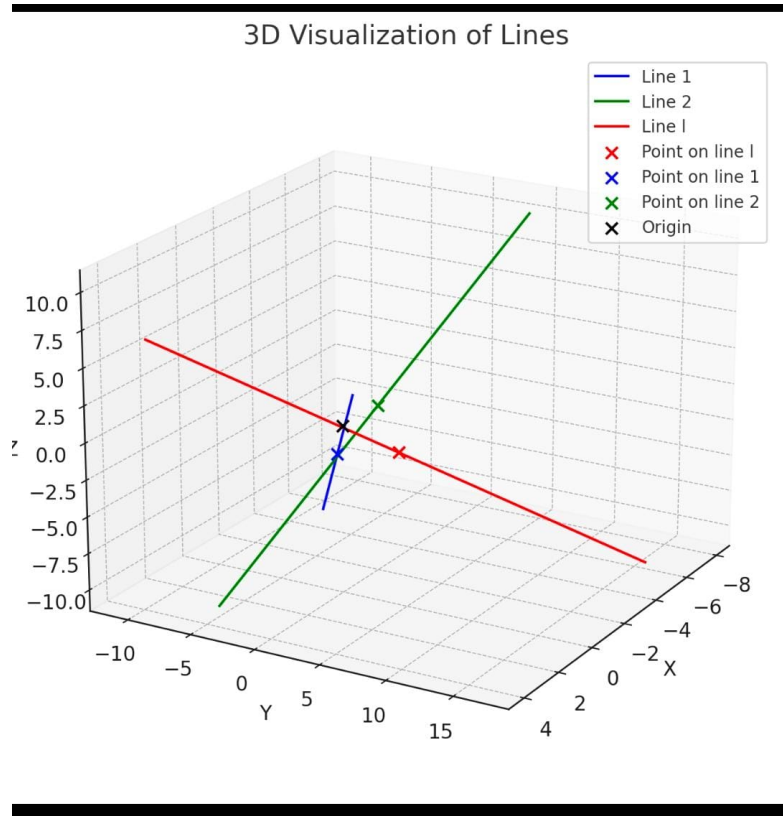


Fig. 0.1